Python Code Testing: Automated 10-Point Unit Test Suite

Phase 6: Unit Test Execution (test_data_integrity.py)

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1. Unit Test Execution Ledger

The automated unit test suite test_data_integrity.py executed with 100% success across all 10 assertion points.

Table 6.1: Automated 10-Point Unit Test Suite Results

Test ID	Test Assertion Name	Target Metric	Status
UT-01	test_01_athletes_file_size_under_25mb	athletes.csv < 25 MB (21.73 MB)	PASS [OK]
UT-02	test_02_athletes_row_count_and_columns	157,417 rows, 18 columns	PASS [OK]
UT-03	test_03_athletes_schema_and_types	Column names and dtypes	PASS [OK]
UT-04	test_04_athletes_medal_values_clean	Medal domain constraints	PASS [OK]
UT-05	test_05_medal_results_row_count	3,837 rows verified	PASS [OK]
UT-06	test_06_medal_results_no_duplicate_keys	Composite key [games_id, noc] uniqueness	PASS [OK]
UT-07	test_07_medal_sum_arithmetic_consistency	Total = G + S + B (0 mismatches)	PASS [OK]
UT-08	test_08_weighted_score_calculation	Score = 3G + 2S + 1B (0 mismatches)	PASS [OK]
UT-09	test_09_usa_medal_tally_verification	USA Gold: 1,131 | Total: 2,823	PASS [OK]
UT-10	test_10_processed_benchmark_files_exist	Processed benchmark CSVs present	PASS [OK]